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The Iris Dataset: An Accidental Case Study in Pedagogy, Methodology and Real-World Implications

"One class is linearly separable from the other 2; the latter are not linearly separable from each other." (UC Irvine Machine Learning Repository , 2025)

* **Author's Note**

Shared with the consultant's permission; identity withheld at the authors' discretion. This case study examines two parallel analytical workflows as expressions of distinct STEM training lineages, not the individuals who performed them. Any consultant-generated artifacts appear only in anonymized form to illustrate methodological contrasts.

1. Introduction

The Iris dataset is ubiquitous in introductory statistics and machine learning. It is typically presented as a parable of clean categorical separation: three distinct species (*I. setosa*, *I. versicolor*, and *I. virginica*) perfectly distinguishable by their measurements. A closer examination reveals a more complicated truth. While *I. setosa* is distinct*, I. versicolor,* and *I. virginica* overlap significantly—a nuance explicitly noted in the dataset's canonical documentation yet frequently erased in introductory pedagogy[6]. The Iris dataset's repository documentation explicitly states that two species overlap, ostensibly quoting from the primary text though no citation is provided; "One class is linearly separable from the other 2; the latter are not linearly separable from each other", yet it is served as the canonical example of perfect classification in statistics and machine learning education (Date: July, 1988).This contradiction is not accidental—it reveals how statistical pedagogy systematically erases empirical complexity to preserve the myth of clean categorical boundaries.

Science is methodology, but methodology is not neutral. The tools we use to observe and interpret the world are shaped by the values and assumptions of their creators. Statistics, in particular, emerged from efforts by a small cohort of White men (non-exhaustive: Galton, Pearson, and Fisher) seeking to validate the racial hierarchies their ideologies insisted upon Clayton, 2020). Creating a measure would shift opinion into the domain of objectivity. Creating measures, scales and standards required a "from" or a point to measure proximity. How else would one know how close one was or was not to the mark? An idealized White, cisgender male body to quantify human "difference" was the benchmark. Everybody non-White, non-cisgender male has been measured against this. Medicine, even today still has its benchmarks (within normal limits) as a white cis male body. Pulse oximetry (Valbuena et al., 2022),

race-corrected eGFR (Inker, L. A et al, 2021; Charles et al., 2024), and pain assessment (Hoffman et al., 2016) are not remnants of some past misstep.

What often goes unexamined is how this logic of deviation from the presumed standard (central tendency) functions as alpha and omega by mirroring and reinforcing systems of social classification and hierarchy. Fisher helped formalize these frameworks from within the institutions of eugenics: he held the Galton Chair of Eugenics at Univeity College London, a chair Galton had endowed, Pearson held it first and the iris analysis appeared in the Annals of Eugenics — the journal he then edited, after being given the position from Galton.

The core research question was as follows: *Are there significant differences between the species, and can they be accurately predicted?* However, the divergence between one analyst's findings conducted through a standard predictive lens (perfect separation) and a second analysis (structural overlap) reframed the inquiry.

Because the consultant and I worked independently, our workflows inadvertently became a natural experiment in statistical pedagogy. One analysis (consultants) followed the predictive pipeline toward separability, while the other (client) used an exploratory-diagnostic lens to surface structural overlap. Their divergence was not a disagreement but the case study itself: a demonstration of how training shapes what analysts see and discard**.**

Through a critical re-analysis, this report demonstrates how the choice of lens determines data visibility. When we prioritize algorithmic separation over diagnostic complexity, we risk creating models that are mathematically accurate but structurally false—a legacy that extends from Fisher's methodology to modern debates on biological determinism in population health, policy, and our lives.

A Note on Positionality and Reflexive Methodology

I approach this dataset through a specific critical lens: one that seeks to uncover how ideological commitments become embedded in seemingly neutral statistical practice. This lens necessarily guides my methodological choices. Unlike the uncritical application of statistical procedures, I explicitly surface my theoretical commitments, following Kincheloe's (2008) notion of "critical transparency." I view the Iris dataset not as a neutral "Hello World" for machine learning, but as a seed crystal of a particular statistical ideology—one that prioritizes separation over overlap, homogeneity over heterogeneity and prediction over understanding.

2. Methods

2.1 Data Source

The Iris dataset is served by the UC Irvine Machine Learning Repository [link]. The dataset contains 150 observations of iris flowers, divided equally among three groups: *I. setosa, I. versicolor*, and *I. virginica*. Four features were measured for each sample: Sepal length, Sepal width, Petal length, and Petal width (UC Irvine Machine Learning Repository, 1988)

2.2 Exploratory Data Analysis (Client)

To establish a way into the data, an Exploratory Data Analysis (EDA) prioritizing discovery and visualization was conducted (EDA here). This included generating summary statistics, pairwise scatterplots to identify multicollinearity, and unsupervised K-Means clustering to test natural grouping without labels (see appendix for tables and secondary figures to reduce clutter in the main document).

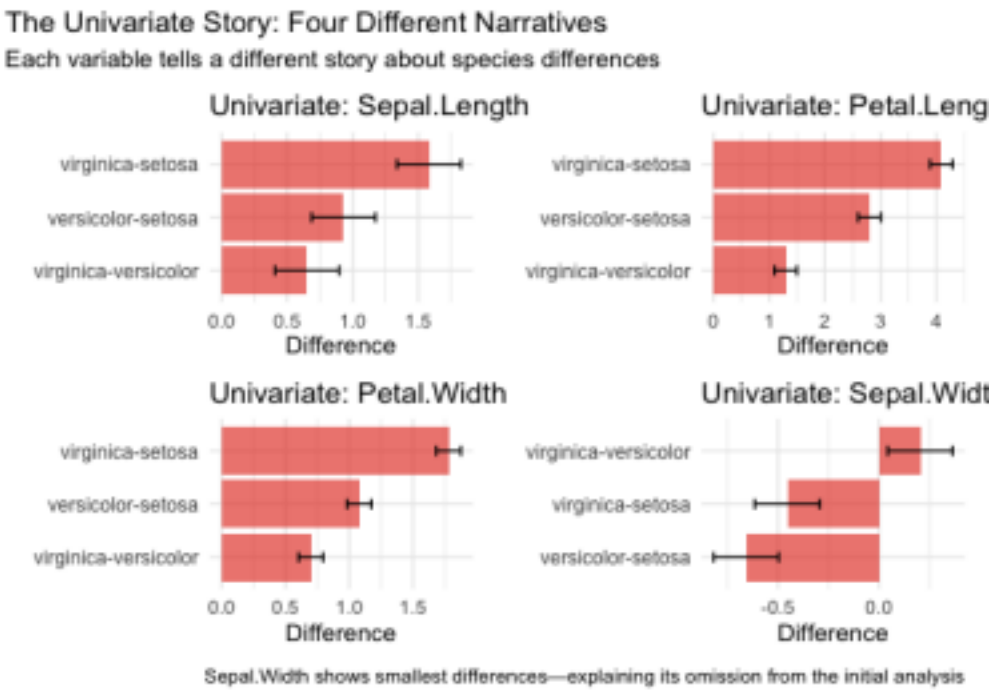


Figure 1. Distribution of Floral Measurements by Species. *The horizontal orientation reveals a critical pattern: consistent whisker lengths across species suggest that within-group variation is similar, differing primarily in central tendency rather than in distribution shape. This visual impression of "shifted" but similarly shaped distributions—rather than fundamentally different types—challenges the strict assumptions of normality and independence required for the consultant's* standard ANOVA approach. Note specifically the substantial overlap in *Sepal* width, which contradicts the hypothesis of clean categorical separation.

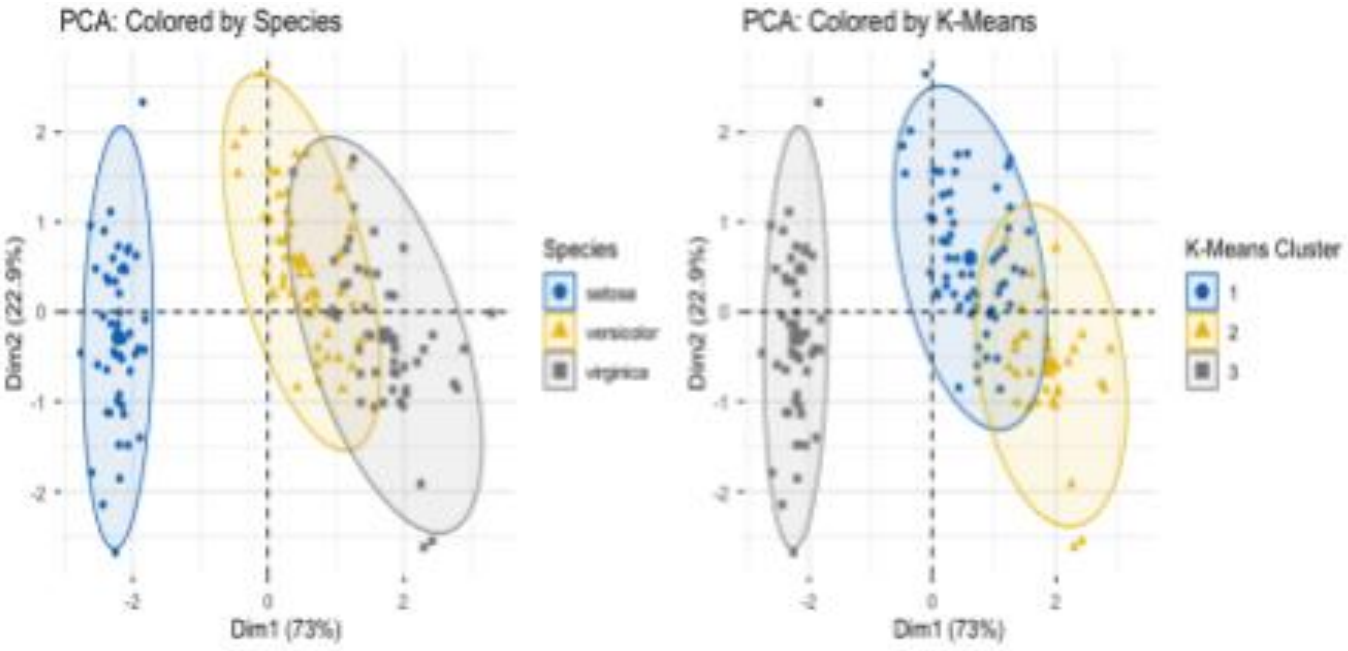


Figure 2. *Principal Component Analysis (PCA) Colored by Species and K-Means Cluster*

Assignment Left panel: PCA of the Iris dataset colored by species (I. setosa, I. versicolor, I. virginica), showing clear separation, particularly along PC1. Right panel: The same PCA plot colored by K-means cluster assignment (k=3), demonstrating approximate correspondence between unsupervised clusters and true species groups. Ellipses indicate 95% confidence intervals for each group (K-means methodology and interpretation follow the resource published by Columbia University Mailman School of Public Health)**.**

2.3 Consultant Analysis

The consultant operated independently drawing from a meeting with the client and left with background information, EDA, research question, and suggested methodologies for consideration in their approach. The consultant's selected methodology is summarized below; the full version is linked here: (Output):

Linear Regression: Three separate linear models were fitted for *Sepal* length, Petal length, and Petal width. Notably, Sepal width was omitted from this testing phase.

Principal Component Analysis (PCA): A second PCA, repeating the one in the client's EDA, was conducted,

followed by a linear regression of the first principal component against species.

Multinomial Logistic Regression: A supervised classification model was fit to predict species based on floral measurements.

2.4 Remedial Analysis (Client)

Upon receiving the consultant's report, the client identified significant methodological gaps and performed the following analyses (Here):

MANOVA: To account for the relationship between dependent variables, replacing the isolated linear regressions.

Tukey's HSD: To provide the necessary post-hoc pairwise comparisons that the consultant omitted.

Linear Discriminant Analysis (LDA): To evaluate the potential separation of classes using a methodology appropriate for the data structure. I would be remiss to not mention the author of this method (R. A. Fisher) and in particular with this dataset (Fisher, 1936). Today we would note this as a potential conflict of interest and invite additional validation to mitigate that potential.

3. Results

The investigation revealed a stark contrast between the illusion of clean separation presented in the consultant's initial report and the nuanced reality uncovered through re-analysis.

3.1 The Consultant's Analysis

The delivered workflow comprised three components. Three separate linear regressions of species on Sepal length, Petal length, and Petal width each returned high significance ($p < 2e\text{-}16$); Sepal width was omitted from this phase. A PCA over all four variables (PC1: 72.96% of variance) was followed by an ANOVA of PC1 scores against species ($F = 1051$, $p < 2e\text{-}16$). A multinomial logistic classifier over all four

features achieved 148/150 on the full dataset (98.7%) and 49/50 on the held-out split (98%), with every error a *I. versicolor* ↔ *I. virginica* confusion. The raw logs show repeated convergence warnings ("fitted probabilities numerically 0 or 1 occurred") absent from the delivered report. Figure 3 displays the consultant's PC1 visualization.

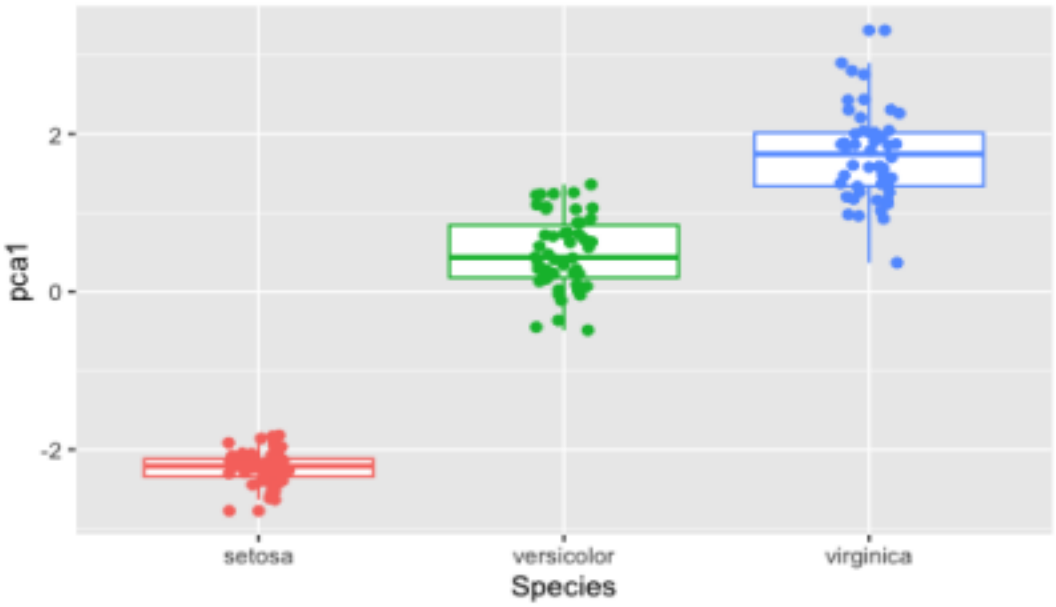


Figure 3. (Consultant Output). Principal Component Analysis (PC1) scores by species. *Principal Component Analysis (PCA) was used to visualize the scores of the first component (PC1) by species. PC1 accounted for 72.96% of the variance and effectively maximized the separation between the groups. This visualization serves as a summary of the primary variation, distinct from the bi-dimensional structural overlap (PC1 vs PC2) identified in the initial Exploratory Data Analysis* (see Appendix)

The consultant ran linear regressions (functionally equivalent to ANOVA) on three of the four variables: Sepal length, petal length, and petal width. All returned high significance ($p < 2e\text{-}16$). However, the consultant excluded Sepal width—the one variable that shows significant overlap between *I. versicolor* and *I. virginica*. While *Sepal* width was retained in the final classifier, it was systematically omitted from the testing phase. Even with sepal width included, the classifier's only errors are *I. versicolor*↔ *I. virginica* confusions; the overlap survives inclusion.

3.2 Client Re-Analysis

The client performed the analyses the consultant's workflow omitted:

The remedial workflow analyzed the complete four-variable feature set. MANOVA showed a significant multivariate effect of species (Pillai's trace, $p < 0.001$). Tukey HSD comparisons were significant for all twelve contrasts; the smallest was *I. virginica–I. versicolor* sepal width (0.204 cm, 95% CI [0.043, 0.365], ($p_{adj} = 0.009$) — the lone comparison above $p < 0.001$, with the greatest distributional overlap. LDA produced a first discriminant axis accounting for 99.1% of between-group variance, on which sepal width carried a coefficient of 1.53 against sepal length's 0.82 (petal dimensions: −2.20, −2.81); accuracy was 98% (147/150), again with all errors on the *I. versicolor – I. virginica* boundary. Figures 4A/4B present the post-hoc intervals and the LDA projection.

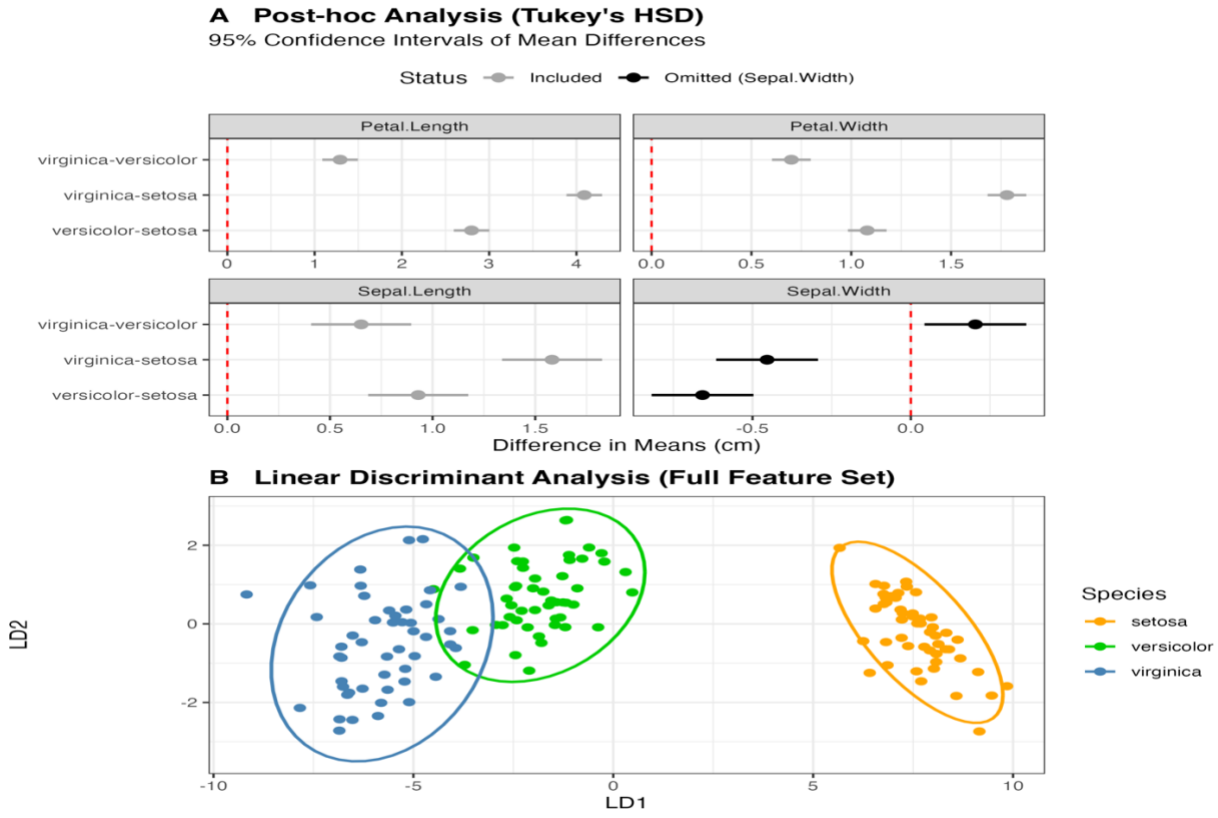


Figure 4A/4B. Reintegration of excluded features reveals significant biological structure

deployed for remedial analysis. (A) Tukey's HSD pairwise comparisons of mean differences for all four morphological traits; bars are 95% confidence intervals. Variables in gray were retained in the consultant's inferential models; Sepal width (black) was omitted from them. Despite its omission, Sepal width shows significant differences for multiple species pairs — the *I. virginica–I. versicolor* interval (0.204 cm, 95% CI [0.043, 0.365], p_adj = 0.009) excludes zero, though by a margin that renders as touching zero at shared scale. (B) Linear Discriminant Analysis using the full four-variable feature set. Ellipses indicate 95% coverage regions (bivariate-normal assumption) for *I. setosa* (orange) *I. versicolor* (green), and *I. virginica* (blue). The latent structure confirms distinct clustering of *I. setosa* along LD1 while honestly depicting the biological proximity and slight overlap between *I. versicolor* and *I. virginica*, a reality obscured by aggressive feature selection. Plots regenerated from source for this version; the generation script accompanies the appendix.

4 Discussion

The consultant's analysis painted a picture of definitive separability, but this conclusion relied on systemic methodological choices.

4.1 The Significance of Omission

Post-hoc tests revealed the dataset's only marginal boundary. The system requires analysts to stop at $p < 0.001$ because deeper interrogation exposes the dataset's explicit warning about overlap. This isn't incompetence; it's the logical endpoint of teaching statistics as 'find significance, then stop'.

Examination of the Linear Discriminant coefficients reveals the magnitude of exclusion errors forcing the model to rely on the weaker signal of Sepal length.

4.2 Conflation of Prediction and Inference

While linear regression and ANOVA are mathematically equivalent (both represent the General Linear Model), the consultant utilized regression purely as a classification tool rather than a hypothesis test. A rigorous ANOVA framework asks, *"Do specific group means differ?"* and mandates assumption checks and post hoc analysis. In contrast, the consultant's approach implicitly asked, *"Can I separate these groups?"* By prioritizing separation over inference, the analysis treated the biological overlap in *Sepal Width* not as a diagnostic finding but as "noise" to be discarded in pursuit of a cleaner, albeit less accurate, model.

4.3 Circular PCA Logic

PCA was applied to a labeled dataset to extract a variance-maximizing summary, which was then plotted against the same labels. Within a descriptive pipeline, this confirms separation by design rather than testing it independently. The overlap visible in PC1 vs. PC2 (EDA) is structurally real; reducing it to a single axis for classification erases the diagnostic signal.

4.4 The Broken Classifier: Accuracy as Separability

Within a predictive pipeline, these warnings are often suppressed as computational noise. Structurally, they signal complete separation driven by *I. setosa,* while the *I. versicolor–I. virginica* boundary remains unresolved. Accuracy here measures separability, not diagnostic validity. This is not merely a statistical shortcut; it is the bedrock of biological determinism/essentialism.

Post-hoc tests would have revealed the dataset's only marginal boundary: on sepal width, *I. versicolor* and *I. virginica* differ by just 0.2 cm ($p_{adj} = 0.009$, the lone pairwise comparison above $p < 0.001$)—statistically detectable, yet corresponding to distributions that overlap almost entirely.
The logistic regression 'succeeds' (148/150, 98.7%) outcome-to-variable association because it treats *I. setosa's* separability as the goal, not a diagnostic, not as a red flag, or not as, perhaps, nothing at all. Sometimes a flower is simply a flower among other flowers.

4.5 Two Coherent Lineages

It is critical to acknowledge that the consultant's approach was not technically 'incorrect' within the paradigm of standard data science training; rather, it was optimized for a different objective. The consultant's methodology prioritizes parsimony and separation, which are the hallmarks of efficient predictive modeling. By this metric, the removal of 'noisy' variables, such as *Sepal Width,* is a feature, not a bug, as it creates a cleaner, more definitive classification boundary.

The client's objective as an equity-centered scientist was not classification, but rather diagnosis and structural understanding. Through this lens, the 'noise' of overlapping groups is not waste to be discarded but the primary signal of biological (or social) reality. The friction in this collaboration did not arise from errors but from the distinct lineages of STEM training we came from: one shaped by the predictive logic of

statistics and the other shaped by the structural and contextual demands of population health. Both approaches were internally coherent; their divergence simply revealed how training directs attention to different forms of truth. The consultant succeeded in building a map that was easy to read, and the client required a map that was true to the terrain.

5. The Canonical Record

The organic banal graduate project episode documented here is paralleled at the level of the field's infrastructure. Almost as if a confirmatory nod about the evidence chain for the unintended hypothesis. It is well established that primary source information is superior to secondary source. A proxy, or representation of information contains some degree of lossiness. Primary source is always preferable. If you have ever played the children's game telephone you are familiar with lossiness. Is the information you give the same information that is reproduced? Primary source material is always first choice barring some significant rationale for not being able to utilize it. It is a wonder then that the repository does not serve the primary source. Fisher's 1936 paper printed the measurements directly; the file the repository distributes instead descends from an unrecorded chain (Appendix D.2 Full provenance documentation — source verification, mirror comparison, and extraction methodology — is provided in Appendix D, available at go.dataacorns.com/the-record). The record in the archive is cavalier stating; "Past Usage: - Publications: too many to mention!!! Here are a few..." And proceeds to list three. This legacy record names a creator and a 1988 donor, and nothing between (Appendix D.1). The primary has been continuously available for ninety years. Its absence from the record is a choice, maintained across a full platform modernization. And passed along generations of statistics, machine learning and computer science students.

Where the primary is explicit, the record reverses it. Fisher's text introduces Table I thus: the *I. setosa* and *I. versicolor* plants were "found growing together in the same colony and measured by Dr E. Anderson, to whom I am indebted for the use of the data" (Fisher, 1936, p. 179). One colony, one measurer, and the analyst describing himself as a borrower of data he did not collect. The canonical record credits Fisher as "Creator" — first in the legacy file, again in the modern API's creator field, and the collector's name appears nowhere in either (Appendix D.1, D.4). The record does not merely omit the primary's attribution; it inverts it. Attribution, citations and provenance matter. It mattered then; it matters now.

Logically, where the primary is silent, the record is silent too. Of the *I. virginica* sample the paper

says nothing: Table I presents it under a species heading, with no collecting origin anywhere in the text. That silence is why the origin of a third of the field's most-used dataset stayed undetermined until Unwin and Kleinman traced it, by matching Fisher's frequency tables against Anderson's published collections, to a single Tennessee colony sampled in 1926 — nine years before the other two species, in a different country, by the same measurer (Unwin & Kleinman, 2021; Anderson, 1935).

The founder knew what curation meant, and said so. Fisher notes that the virginica sample "differs from the two other samples in not being taken from the same natural colony as they were — a circumstance which might considerably disturb both the mean values and their variabilities" (Fisher, 1936 p. 185). The caveat has sat in the founding paper since 1936. It survives in the one document the record does not serve — and in none of the documents it does.

The record does not even serve the primary's data. The record's own erratum identifies three cells in which the distributed files differ from Fisher's printed table (Appendix D.1); I verified the primary's values against the printed scan directly (Appendix D.6). The erratum is accurate, displayed verbatim on the modern dataset page, and unapplied: the canonical download still serves the deviant values, directly beneath the text that corrects them (Appendix D.1, D.2). Why list a fix and not apply it? That implies the error was innate and it is not. As if the goal was archival integrity. No such error exists in the original. The consequence is not cosmetic. In the canonical file, the two corrupted *I. setosa* specimens collapse onto a third, forming a triplet of identical rows; a standard deduplication deletes the two corrupted specimens and the one genuine duplicate — reducing n from 150 to 147, the I*. setosa* count from 50 to 48, and the I. virginica count from 50 to 49 (Appendix D.3). The same operation on R's built-in copy — which matches the corrected lineage — deletes one row. And the duplicate the record does share with the primary is genuine: Fisher's own table prints one virginica specimen twice, so the celebrated 150 observations have numbered 149 plants since 1936 (Appendix D.6). A rounding error? Convenience? The original dataset is not wrong; we wanted it to be something it was not. Mirror choice is an unrecorded analytical variable on the field's most-analyzed dataset, and no analysis states which mirror it used. The Iris dataset is taught as clean, easily classed data and it never was. It has been cherry picked, and manipulated to fit a story. A storyline written by its own founders. While ironic that it suffered the same fate, it would be illogical to say it is unexpected. The need we have created of forcing results, model output and linear separability has created an inaccurate (dishonest) environment where analysts, students, researchers and teachers are not rewarded for checking their work but instead, results, outputs and deliverables matter more. And this is in part how, copying a 600-cell table is more expensive, over the course of a morning than not doing so, over nearly a century. Linear separability has demonstrable success given our division.

The pattern of disclosure without solutions is extensive to say the least in this corpus. The equivalent of; " I'm keeping it real" by telling you how I feel but truthfully, it is a justification to abdicate responsibility and simply vent feelings. These lazy patterns are of the same lineage. Where convenience matters more than outcomes, where data is a toy for training something else, a mere stepping stone. Lest we forget data is me, you and everything that matters to us. The consultant's chunks carried warnings his options hid; the legacy record documents its own corruption in a note that misspells "discrepancy," gives a date without a year, and misspells the journal that published the founding paper ("Annual Eugenics"); the modern page prints the erratum beside the uncorrected file; the modern metadata carries a truncated DOI for its own introductory paper and dates the dataset's creation to 1936 — the year of the analysis, a full decade after the first collection (Appendix D.4). Three interventions in ~forty years— an update, an erratum, a platform migration and no fixes. The single exception instructs: the help page of a statistical programming environment, volunteer-maintained, carries the correct journal name, a working DOI to the founding paper, and a citation crediting Anderson with the collection. Provenance did not survive in the archive; it survived in the ecosystem. The knowledge of what this data is, was never lost. As per our collective attention demonstrates, it is simply not where we chose to look; at one and a half million views and counting (Appendix D.4). The world's most analyzed dataset has errors throughout, no one checked, fixed, or implemented for ninety years.

I reported these findings to the repository maintainers on September 13, 2026 (issue #32); the served record remained unchanged as of September 22, 2026 (Appendix D.5).

In biology, *I. versicolor* and *I. virginica* are overlapping taxa. The founding paper said as much; the record dropped it; this project re-enacted the drop. The episode analyzed in this case study is not an anomaly but the pedagogy's ordinary operation, observed at small scale, on a dataset whose canonical record performs the same operation at institutional scale. Curation hides this. The repeated, systemic use of this dataset, presented in this way, accomplishes the same. The fan out, amplified over nearly a century is unquantifiable.

6. Conclusion

In population research, this produces race-based models that "predict" disparities while ignoring structural drivers. By asking *"how can I separate groups?"* (classification) rather than *"why do these groups*

differ?" (inference), analysts encourage the use of race as a biological proxy for outcomes. A costly conflation.

We see this logic fail in high-stakes contexts, such as using race to predict infant mortality while ignoring social determinants (Bishop-Royse et al., 2021), and in everyday policy, such as the debate in Wisconsin over when older adults should stop shoveling snow (American Heart Association, 2026).

By treating age as a binary cutoff for "shoveling ability," we ignore functional health. By treating race as a proxied risk factor for health, we ignore racism (Garvick et al., 2023). In both cases, the model creates a dangerous fiction: it reifies the category (age/race) as the cause, effectively hiding the actual mechanism (health/racism). This protects the system from scrutiny while pathologizing the individuals within.

This case study demonstrates that the Iris dataset is not a neutral teaching tool; rather, it is a seed crystal for a statistical ideology that prioritizes separation over structural reality. The client and the consultant became a textbook example of this legacy in 2025, a convergence so precise that it could not have been improved by orchestration. Its lack of orchestration is what makes it so revealing. Moving forward requires training analysts not just to run code but to recognize when a proxy is masking a mechanism. We must learn to retain the Sepal widths—the messy, overlapping variables—because that is often where the structural truth resides. Or at least, where the useful information is.

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